CURBCOUNT · TRACEABILITY MATRIX

CurbCount traceability matrix

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/curbcount		
Source	docs/03b-traceability.md @ 82b4850		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-31	Amish Chadha	Generated by .kit/trace.py
0.2	2026-10-31	Amish Chadha	Generated by .kit/trace.py
0.1	2026-10-31	Amish Chadha	Generated by .kit/trace.py

CurbCount traceability matrix

Generated by `python .kit/trace.py` from the user requirements, product requirements, calculations, build plan and BOM. Do not edit by hand. 16 user requirements, 21 product requirements, 0 gaps.

Table 1. User requirement to product requirement to evidence to verification.

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-003	PR-001	Count three classes in two directions	CBC-CAL-001, CBC-BLD-001, CBC-STM-001, CBC-OPT-001	Analysis (CAL-001 C), TRL 3; test against manual counts, TRL 4	At risk: 4.2 frames for a point at road level, 3.2 for a warm hood 1.0 m up (STM S5), against 4 assumed needed; fix proposed to Amish (CBC-DEC-001 O3)
UR-004	PR-002	Pedestrian and cyclist accuracy	CBC-CAL-001, CBC-OPT-001	Analysis (CAL-001 C), TRL 3; field comparison with manual counts, TRL 4	At risk: random merging alone loses 6.5 %; groups add more
UR-004	PR-003	Motor vehicle accuracy	CBC-CAL-001	Analysis (CAL-001 C), TRL 3; field comparison, TRL 4	To verify at TRL 4: 4.2 frames per car, 3.4 px smear
UR-006	PR-004	Sensor cannot resolve a face or a plate	CBC-CAL-001, CBC-BLD-001, CBC-STM-001	Analysis (CAL-001 B), TRL 3; design review and firmware audit, TRL 4	Met on paper: 7.6 px/m (8.5 for a 2.0 m person) for passers-by; an installer within 0.5 m is above the detection level
UR-007	PR-005	Open 10-byte count record	CBC-CAL-001, CBC-BLD-001	Inspection (design review, CAL-001 G), TRL 3	Met by design
UR-005	PR-006	Coverage from one pole	CBC-CAL-001, CBC-BLD-001	Analysis (CAL-001 A), TRL 3; field check, TRL 4	Met on paper with the array turned 90 degrees: -4.45 to 8.34 m across
UR-009	PR-007	Survive -20 to +50 degrees C	CBC-CAL-001	Inspection (part ratings, STM S7, S8), TRL 3; thermal chamber, TRL 4	To verify at TRL 4: cell discharge rating and window film not yet measured
UR-009	PR-008	Count in sun and shade at temperate sites	CBC-CAL-001, CBC-STM-001	Analysis (CAL-001 D, STM S3), TRL 3; seasonal field test, TRL 4	At risk: on the mild design day sunlit pavement hides pedestrians for 10.2 h with single frames, 3.2 h with track averaging; shaded 0 h. Summer days above 20 degrees C are outside the claim (the independent review put a 15 to 28 degrees C day at about 12 h blind on shaded pavement; not recomputed here)
UR-010	PR-009	Three days without sun	CBC-CAL-001, CBC-BLD-001	Analysis (CAL-001 E2), TRL 3; bench test, TRL 4	Met on paper: 6.22 days on one cell (4.36 at -20 degrees C, 3.49 with the end-of-life 80 % as well)

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-010	PR-010	Energy neutral in winter	CBC-CAL-001, CBC-BLD-001, CBC-STM-001	Analysis (CAL-001 E3, STM S6 and S8), TRL 3; field log, TRL 4	At risk below about 0 degrees C air (cell may not charge below 0 degrees C; CBC-DEC-001 O1). Met on paper above it: 4.4 Wh/day stored against 2.47 Wh/day
UR-011	PR-011	IP65 enclosure and head	CBC-CAL-001, CBC-BLD-001	Inspection (design review, CAL-001 I), TRL 3; spray test, TRL 4	To verify at TRL 4
UR-002	PR-012	Mass on the pole	CBC-CAL-001, CBC-BLD-001	Analysis (CAL-001 H5, cad/fidelity.json), TRL 3; weighing, TRL 4	Met on paper: 5.24 kg with bought parts (CAD model 4.8 kg)
UR-001	PR-013	Install in 60 minutes by two people	CBC-CAL-001, CBC-BLD-001, CBC-STM-001	Analysis (CAL-001 H6, BLD-001), TRL 3; timed trial, TRL 4	To verify at TRL 4: 59 min estimate for the task list in CAL-001 H6; it leaves out platform set-up, traffic control and teardown, which the independent review put at 75 to 90 min in all
UR-001	PR-014	No drilling, welding or pole wiring	CBC-CAL-001, CBC-BLD-001	Inspection (BLD-001, BOM lines 5 to 7), TRL 3	Met by design
UR-015	PR-015	Withstand 35 m/s gusts	CBC-CAL-001, CBC-BLD-001	Analysis (CAL-001 H1, H3), TRL 3; bracket calculation, TRL 4	Met on paper: panel 52 N, post factor 55, twist factor 3.3
UR-013	PR-016	Parts cost within the value-engineering target	CBC-CAL-001, CBC-STM-001	Analysis (BOM, CAL-001 J1), TRL 3	At risk: 274.00 USD is 1.00 USD (0.4 %) under the target, less than the uncertainty in indicative prices
UR-011, UR-012	PR-017	Five years outdoors, one cell change	CBC-CAL-001	Design review (CAL-001 I, STM S18 and S19), TRL 3; ageing and field trial, TRL 4	To verify at TRL 4: window film grade and joint coatings not yet specified
UR-014	PR-018	Radio within licence-free limits	CBC-CAL-001, CBC-BLD-001	Analysis (CAL-001 G), TRL 3	Met on paper: 0.82 s/h at SF9 (limit 36 s/h); 371 ms at US915 DR0
UR-016	PR-019	Public notice plate	CBC-CAL-001, CBC-BLD-001	Inspection (BOM line 12, BLD-001), TRL 3	Met by design (CBC-DDR-001 D7)

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-008	PR-020	Aim check on site	CBC-CAL-001, CBC-BLD-001, CBC-STM-001	Analysis (CAL-001 H6), TRL 3; demonstration, TRL 4	Met by design: 8 min estimate
UR-015	PR-021	Panel and head retained by lanyards	CBC-CAL-001, CBC-BLD-001	Inspection (BLD-001, BOM line 13), TRL 3	Met by design

Gaps

None.